

A Countable-Capture Obstruction for the Ball-Covering Property

Run fa_banach_001, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1 (countable-capture obstruction). *Let X be a normed space. Suppose that for every countable set $C \subset X$ there are a normed space Y_C , a contraction*

$$E_C : X \rightarrow Y_C,$$

and an element $u_C \in S_X$ such that

$$E_C u_C = 0 \quad \text{and} \quad \|E_C c\| = \|c\| \quad (c \in C).$$

Then X fails the ball-covering property.

Corollary 1 (C^* -algebra visibility form). *Let A be a C^* -algebra. Assume that for every countable $C \subset A$ there are a contraction $E_C : A \rightarrow Y_C$ into a normed space and a unit vector $u_C \in A$ such that $E_C u_C = 0$ and $\|E_C c\| = \|c\|$ for all $c \in C$. Then A fails BCP.*

In particular, the conclusion holds if the maps E_C can always be chosen as contractive projections or conditional expectations onto subspaces or subalgebras which norm the proposed centres and whose kernels contain a unit vector.

Idea of the Proof

BCP is countable, so one is allowed to inspect the proposed centres before building an escaping unit vector. The hypothesis gives a contractive “shadow” map which keeps the norms of all proposed centres but sends a fresh unit vector to zero. Applying the shadow map to $u_C - x_n$ forces

$$\|u_C - x_n\| \geq \|E_C(u_C - x_n)\| = \|E_C x_n\| = \|x_n\|.$$

Thus the unit vector u_C lies outside every ball $B(x_n, \|x_n\|)$.

Proof

We use the standard equivalent BCP test for normed spaces: X has BCP if and only if there are countably many centres $x_n \in X$ such that

$$S_X \subseteq \bigcup_n B(x_n, \|x_n\|),$$

where $B(x, r)$ denotes the open ball of radius r around x . This is the form used throughout the source paper and the run packets for arXiv:2311.14257.

Assume the hypotheses of Theorem 1, and let $(x_n) \subset X$ be an arbitrary countable family of proposed centres. Apply the countable-capture hypothesis to

$$C = \{x_n : n \geq 1\}.$$

We obtain a contraction $E_C : X \rightarrow Y_C$ and a unit vector $u_C \in S_X$ such that $E_C u_C = 0$ and $\|E_C x_n\| = \|x_n\|$ for every n . Therefore, for each n ,

$$\|u_C - x_n\| \geq \|E_C(u_C - x_n)\| = \|-E_C x_n\| = \|x_n\|.$$

Hence $u_C \notin B(x_n, \|x_n\|)$ for every n . Since the countable family (x_n) was arbitrary, no countable BCP cover exists. Thus X fails BCP.

Corollary 1 is the same theorem applied to the underlying Banach space of A .

Relation to the Reduced Group C*-Algebra Packet

The partial packet

`2311.14257_reduced_group_cstar_bcp_countable`

is an exact instance of this criterion. If G is an uncountable discrete group and $(a_n) \subset C_r^*(G)$, each a_n belongs to $C_r^*(H)$ for one common countable subgroup $H \leq G$. Choose $g \in G \setminus H$. The canonical subgroup expectation

$$E_H : C_r^*(G) \rightarrow C_r^*(H)$$

fixes every a_n and kills the fresh unitary λ_g . Thus Theorem 1 yields failure of BCP for $C_r^*(G)$.

Scope and Limitations

This is a negative criterion, not a classification of all non-Type-I C^* -algebras. It explains what a future broad non-Type-I obstruction must supply: after countably many proposed centres are known, one needs a contractive map that preserves their norms while annihilating a fresh unit vector.

The condition is not automatic for non-Type-I algebras. Separable C^* -algebras have UBCP, and the source paper gives positive non-Type-I von Neumann algebra cases. Thus an arbitrary non-Type-I classification must distinguish between norm-visible countable-capture mechanisms and genuinely positive nonseparable behaviour.

Novelty and Search Bounds

The local run indexes were searched for arXiv:2311.14257, countable capture, conditional expectations, reduced group C^* -algebras, diagonal escape, BCP, and UBCP. Web searches on June 17, 2026 for the source problem, reduced group C^* -algebras, and the ball-covering property found the source paper and nearby C^* -algebra/operator-space literature, but no exact statement of this abstract countable-capture obstruction. The theorem is an agent-produced abstraction of the reduced-group proof.

Human Review Focus

Reviewers should check:

1. the use of the standard BCP equivalent formulation $S_X \subseteq \bigcup_n B(x_n, \|x_n\|)$;
2. that equality $\|E_C c\| = \|c\|$ for the proposed centres is enough;
3. the C^* -algebra specialization, especially when E_C is supplied by conditional expectations or quotient/capture maps.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Run `fa_banach_001`, lane 19, *Reduced Group C^* -Algebras and the Ball-Covering Property*, solution packet `2311.14257_reduced_group_cstar_bcp_countable`, 2026.
- [3] Run `fa_banach_001`, *Adaptive Diagonal Criteria for Failure of BCP*, solution packet `2311.14257_adaptive_diagonal_bcp_criterion`, 2026.